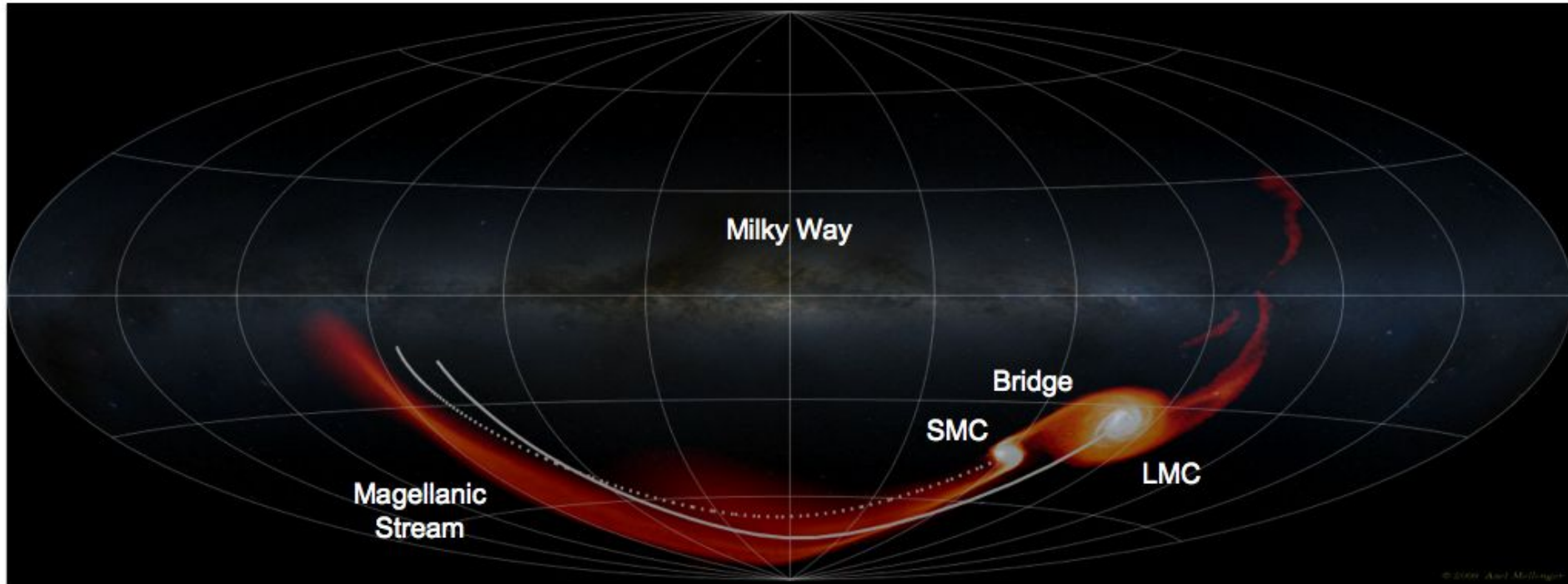


An OGLE/2MASS/Gaia map of the Large & Small Magellanic Clouds

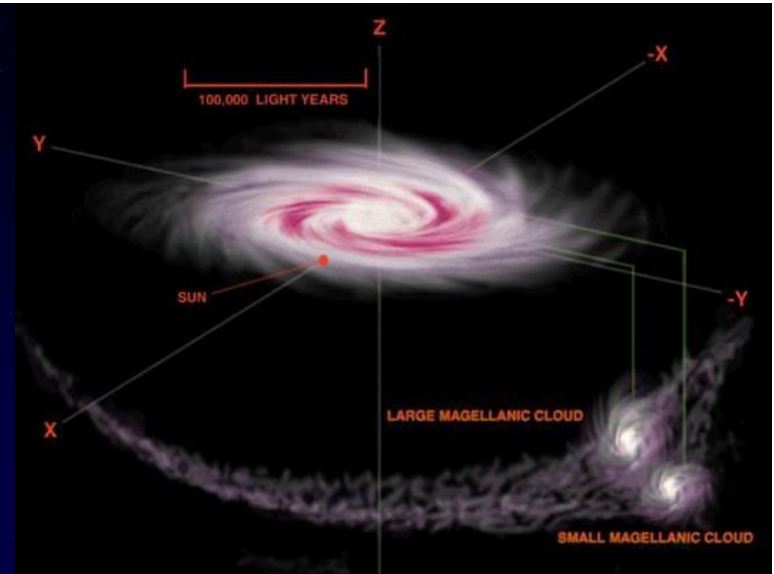
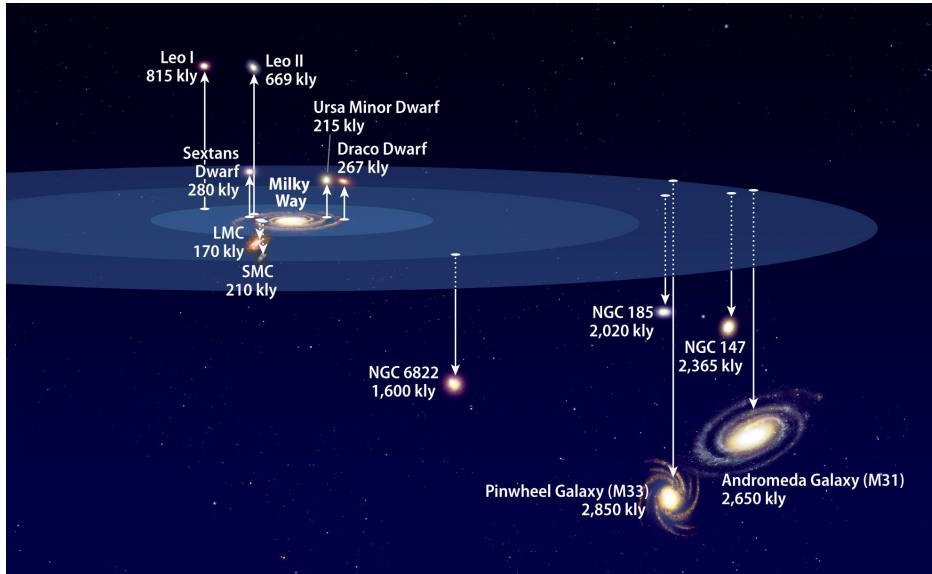
Jack McGuire - Astro 4200 - Fall 2025



The LMC / SMC?

SMC d = 62 kpc
LMC d = 50 kpc

Nearest members of our Local Group to our Milky Way Galaxy. Combined mass of both is 1/10th of the MW. Located along & is the cause of the Magellanic Stream (gas + stars from Magellanic clouds favoring orbit around the MW)



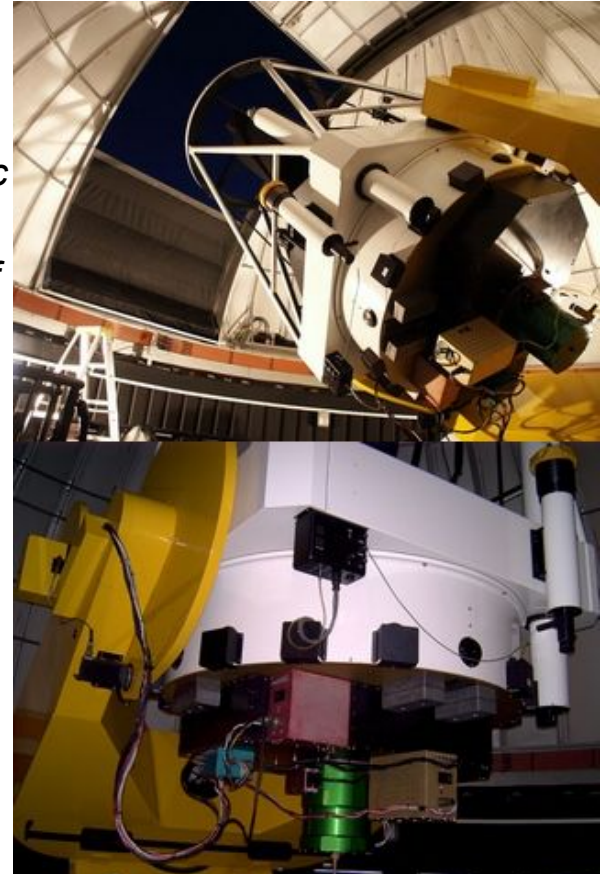
OGLE - Optical Gravitational Lensing Experiment

1.3 m Warsaw University Telescope
Las Campanas Observatory, Chile



While monitoring every star in the SMC + LMC for lensing effects, they tracked the intrinsic variability to the stars, cataloging thousands of variables within the Magellanic clouds, recording periods and mean apparent magnitudes in I & V bands.

- "Third generation" mosaic camera (OGLE-IV, 2009-...)
 - 32 thin E2V44-82 2048×4096 CCD chips



So.. what are we working with?

OGLE Variable Catalog in each object:

- Measured and reports periods and V&I mags for each!

Galactic Bulge & Disk

- 24,000 del scuti
- 78,000 rr lyrae total
- 13,000 spotted stars
- 450,000 eclipsing binaries

LMC Cepheids: 2, 477

SMC Cepheids: 2, 755

RR Lyrae in the LMC: 41, 471

& in SMC: 6, 835

classical Cepheids

- Galactic Bulge
 - OGLE-IV [data download](#) | [paper 1](#) | [paper 2](#) | [paper 3](#)
 - OGLE-III [data download](#) | [paper](#)
- Galactic Disk
 - OGLE-IV [data download](#) | [paper 1](#) | [paper 2](#)
 - OGLE-III [data download](#) | [paper](#)
- LMC
 - OGLE-IV [data download](#) | [paper 1](#) | [paper 2](#)
 - OGLE-III [data download](#) | [paper](#)
- SMC
 - OGLE-IV [data download](#) | [paper 1](#) | [paper 2](#)
 - OGLE-III [data download](#) | [paper](#)
- Dwarf galaxy IC1613
 - OGLE-II [data download](#) | [paper](#)

RR Lyrae stars

- Galactic Bulge
 - OGLE-IV [data download](#) | [paper](#)
 - OGLE-III [data download](#) | [paper](#)
- Galactic Disk
 - OGLE-III [data download](#) | [paper](#)
- LMC
 - OGLE-IV [data download](#) | [paper](#)
 - OGLE-III [data download](#) | [paper](#)
- SMC
 - OGLE-IV [data download](#) | [paper](#)
 - OGLE-III [data download](#) | [paper](#)

So.. what are we working with?

Period-Luminosity relationships

Specific PL relation coefficients in each band, for each object of interest + more.

Table 7. Period-luminosity relations for the three different samples and for each band the form $M = \alpha(\log P - 1.18) + \beta$.

Band	Galaxy	α	$\sigma(\alpha)$	β	$\sigma(\beta)$	Std.dev.
M_K	SMC	-3.179	0.141	-6.081	0.046	0.24
M_K	LMC	-3.282	0.087	-6.225	0.036	0.21
M_K	MW	-3.258	0.092	-6.268	0.030	0.23
M_J	SMC	-2.856	0.169	-5.612	0.055	0.29
M_J	LMC	-3.220	0.090	-5.750	0.037	0.21
M_J	MW	-3.114	0.092	-5.802	0.030	0.23
W_{JK}	SMC	-3.401	0.133	-6.404	0.043	0.23
W_{JK}	LMC	-3.324	0.089	-6.552	0.036	0.21
W_{JK}	MW	-3.357	0.097	-6.589	0.032	0.24
M_V	SMC	-2.705	0.177	-4.238	0.057	0.30
M_V	LMC	-2.775	0.111	-4.499	0.045	0.26
M_V	MW	-2.615	0.100	-4.457	0.033	0.25
M_I	SMC	-2.934	0.156	-5.064	0.051	0.27
M_I	LMC	-3.021	0.101	-5.280	0.041	0.21
M_I	MW	-2.664	0.098	-5.293	0.031	0.21
W_{VI}	SMC	-3.287	0.148	-6.338	0.048	0.25
W_{VI}	LMC	-3.411	0.112	-6.484	0.046	0.24
W_{VI}	MW	-3.084	0.117	-6.562	0.037	0.25
			mag	mag	mag	

Not specific relation for LMC & SMC.
This is the best fit for *all* RR Lyrae

$$M_I = 0.471 - 1.132 \log P + 0.205 \log Z,$$

$$M_J = -0.141 - 1.773 \log P + 0.190 \log Z,$$

$$M_H = -0.551 - 2.313 \log P + 0.178 \log Z,$$

$$M_K = -0.597 - 2.353 \log P + 0.175 \log Z,$$

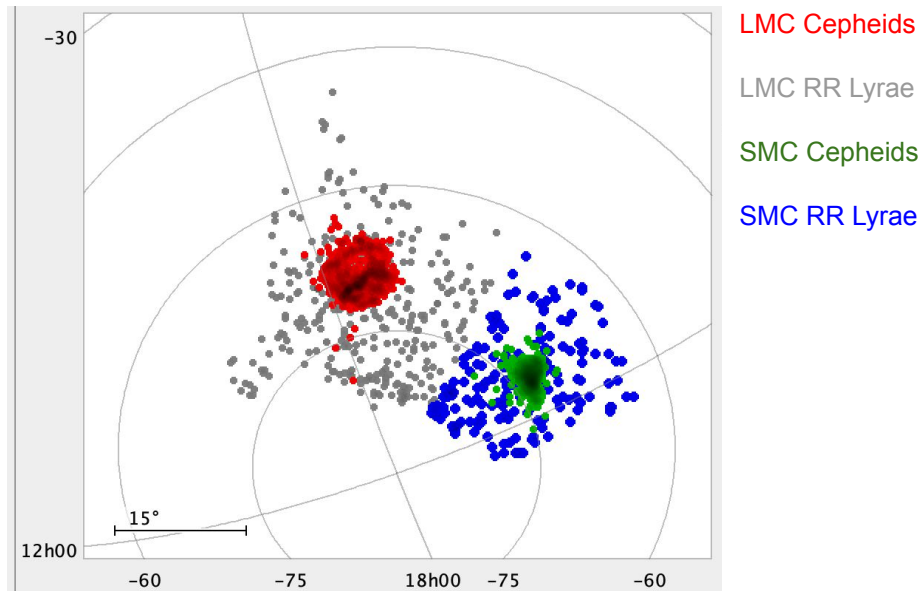
$$M_V = 2.288 + 0.882 \log Z + 0.108(\log Z)^2.$$

$$\log Z = [M/H] - 1.765.$$

$$\text{LMC } -1.5 \text{ M/H} \rightarrow -3.265 \log Z$$

$$\text{SMC } -1.8 \text{ M/H} \rightarrow -3.565 \log Z$$

Initial Sky Plot.



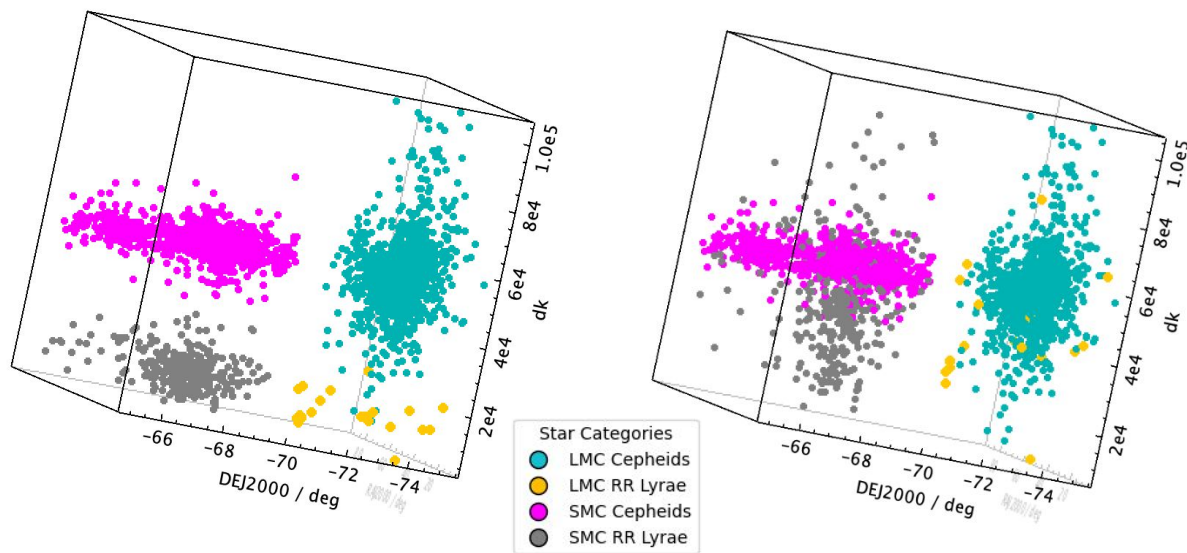
Next was cross-match with 2MASS for H,J,K band apparent magnitudes

(J&K shared between the Cepheid + RR Lyrae relations I found, but *H* is solely for the RR Lyraes)

- Calculate VIHJK for each: absolute magnitude, distance modulus, and distance.

Calibrating the generalized RR Lyrae PLZ relation by including an offset to match the distance of Cepheids.

Adjusted the + or - value until the distances were visibly in line with Cepheids (IN EACH BAND). Since no comparison for H existed, I adjusted H offset until the mean of the mean distances matched roughly the H distance.

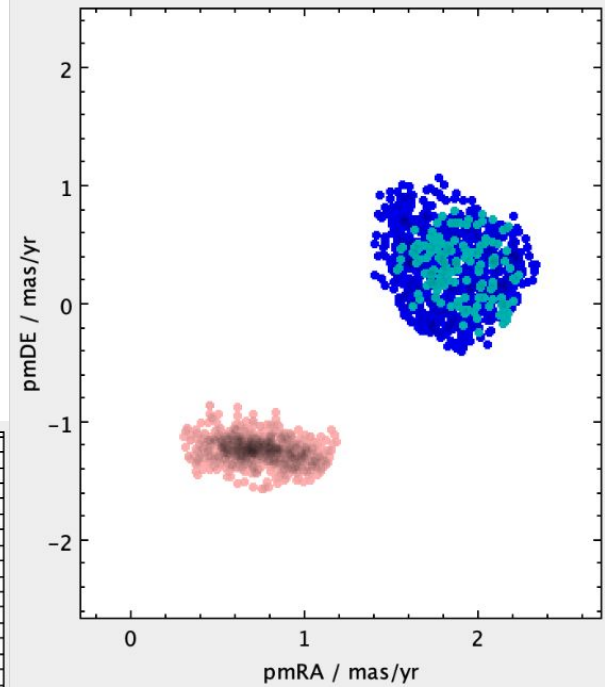
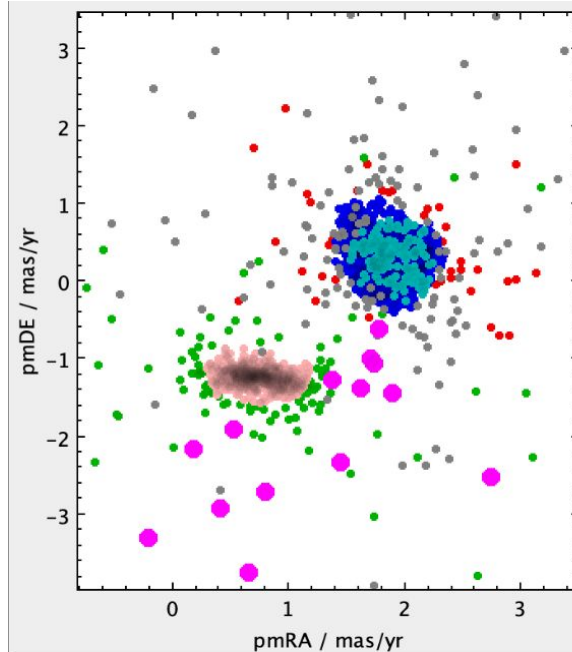


Gaia crossmatch

Parallax gave negative mean, meaningless. Should be ~ 0.02 mas if 50-60 kpc away.

The Proper Motion is much easier to measure and gives slightly more trustworthy results.

No SMC RR Lyraes (in pink) left after crossmatching OGLE with 2mass with Gaia had PMs within the acceptable range of pmDe/pmRA as the rest of the galaxy, so were removed.



Distance modulus (m-M) removal of outliers

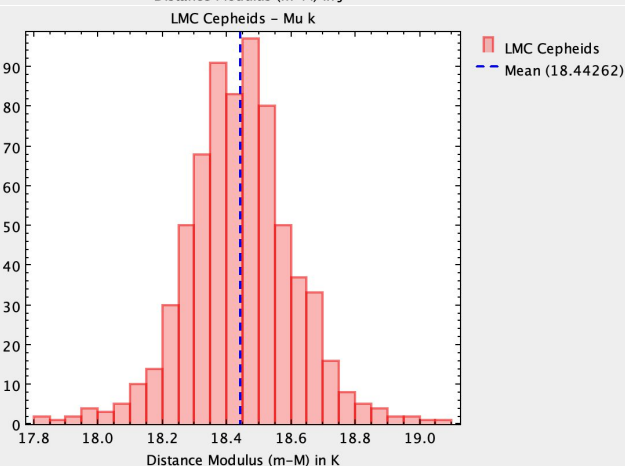
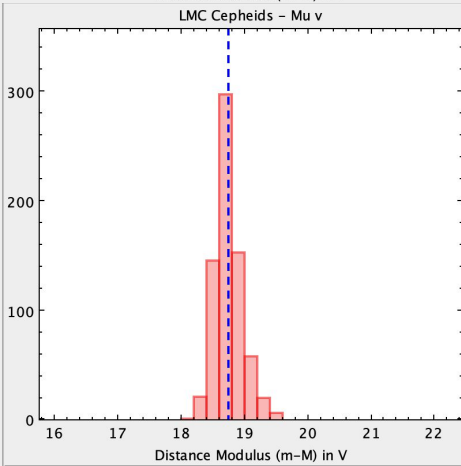
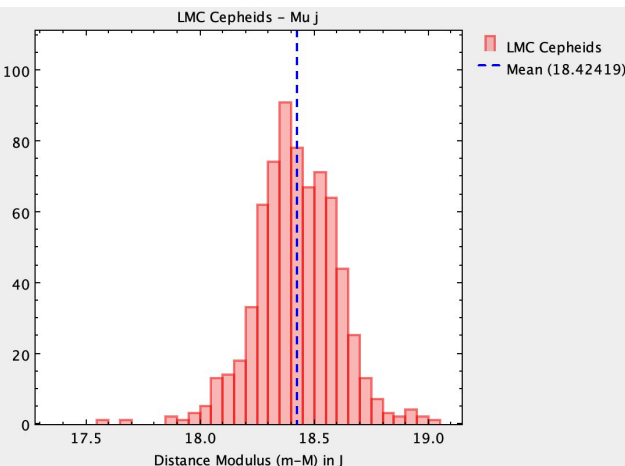
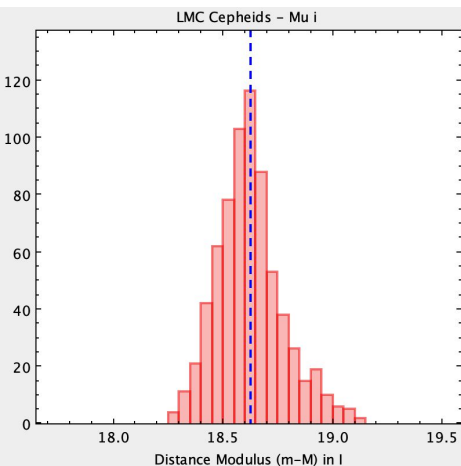
Only kept the stars that had no outliers in any m-M band (IVJKH)

So.. what are we working with?

	After 2MASS match + pm cut	After mu cut
Fundamental Cepheids in LMC:	2, 477 ->	757 -> 699
SMC:	2, 755 ->	827 -> 689
RR Lyrae in LMC (only AB/C selected):	41, 471 ->	138 -> 102
& in SMC:	6, 835 ->	0 -> 0

LMC Cepheid

- means lie 0.1 $\rightarrow$ 0.2 from a μ of 18.5, but close. 18.5 would give ~ 50 kpc



Name	Mean	STD
μ_i	18.62265	0.149951
μ_v	18.74741	0.215976
μ_j	18.42419	0.175375
μ_k	18.44262	0.167758
di	53159.39243	3753.76
dv	56451.61753	5816.65
dj	48556.66428	3909.78
dk	48957.64921	3790.43

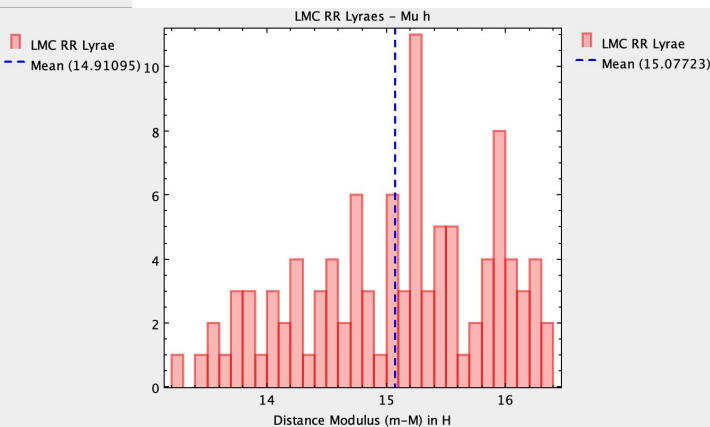
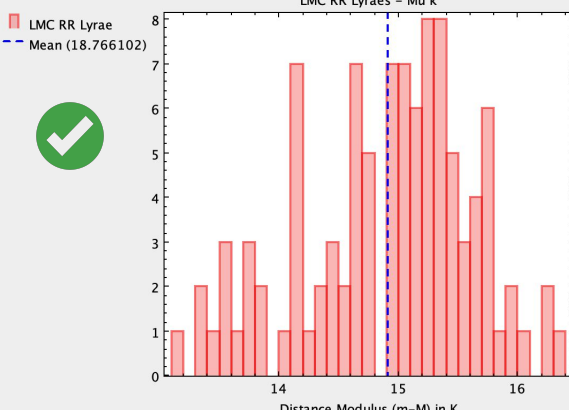
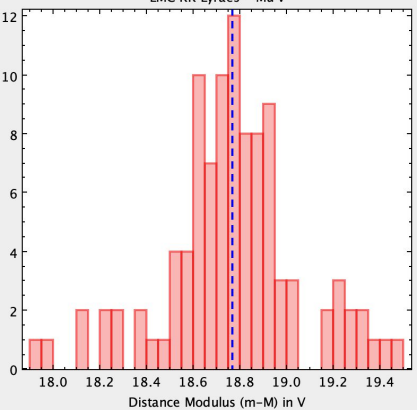
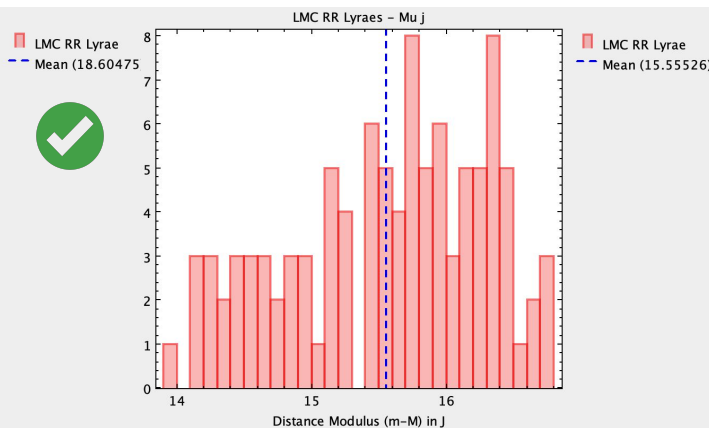
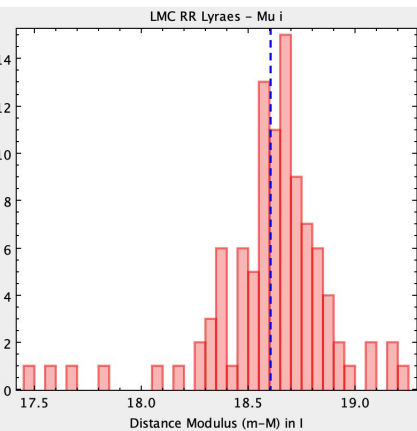
The cepheid relations with the OGLE/2mass periods and photometry yield $\sim$ literature values.



LMC RR Lyrae

General relation

a mu of 18.5 would give ~50kpc



Name

Mean

STD



mu i

18.60475

0.280092



mu v

18.7661

0.289996



mu J

15.55526

0.728881



mu k

14.91095

0.72292



mu h

15.07723

0.786701



di

53011.17004

6358.07



dv

57156.59765

7604.71



dj

13621.4472

4251.27



dk

10117.06537

3175.99



dh

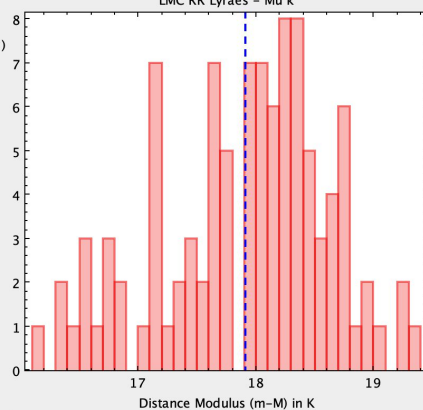
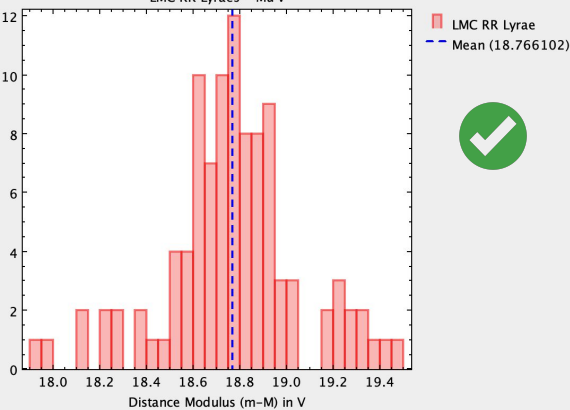
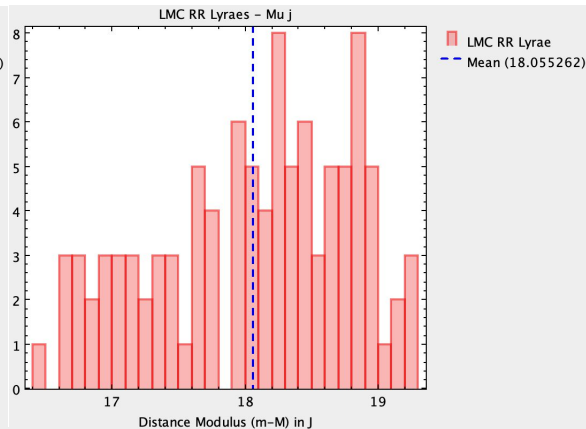
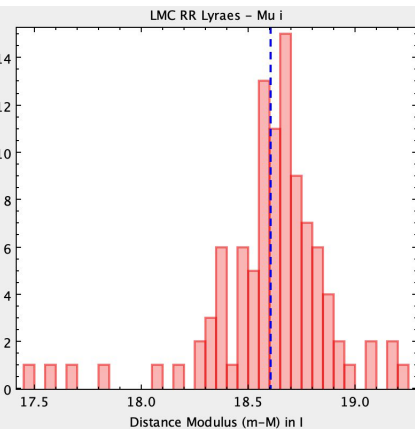
11027.91739

3739.22

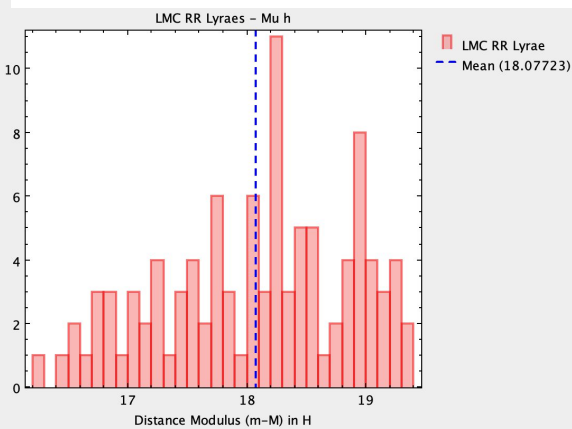
LMC RR Lyrae

a mu of 18.5 would give ~50kpc

With offset adjustment

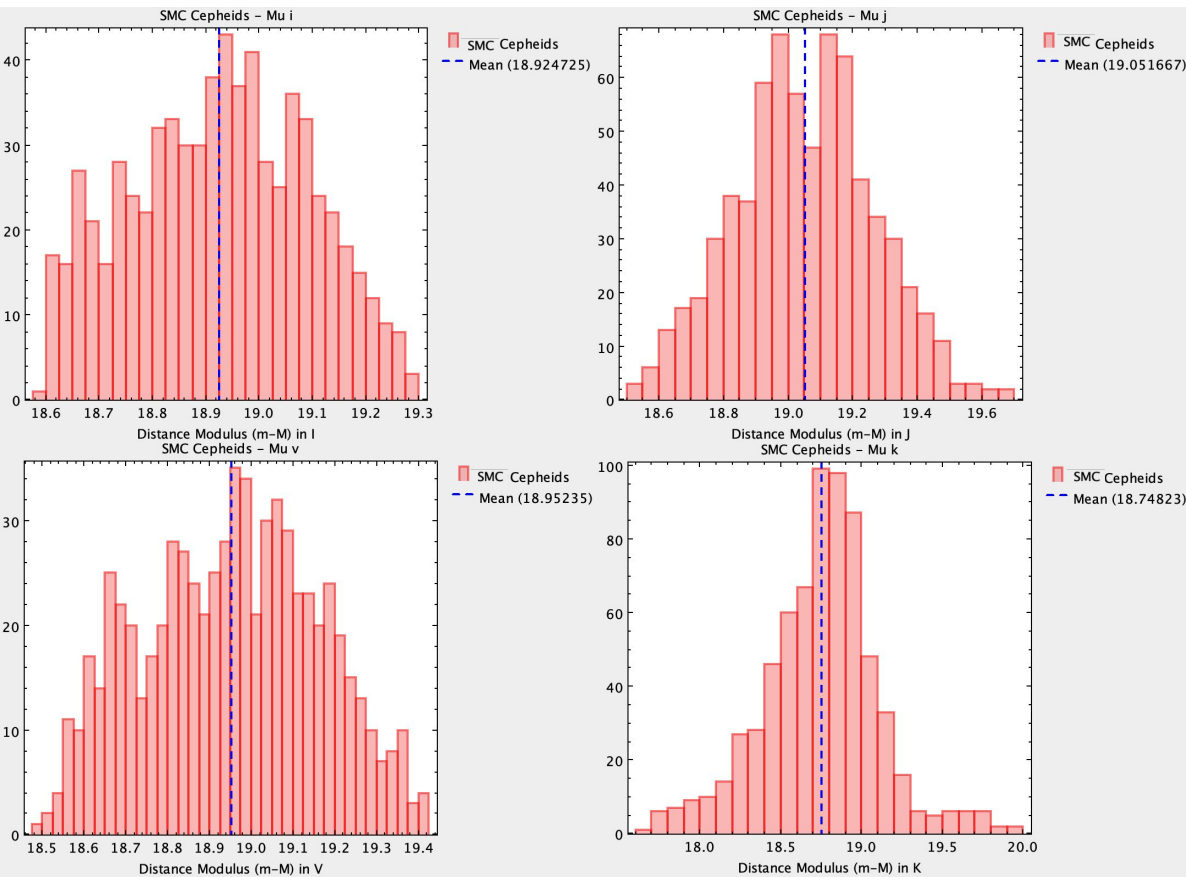


Name	Mean	STD
mu i	18.60475	0.280092
mu v	18.7661	0.289996
mu J	18.05526	0.728881
mu k	17.91095	0.72292
mu h	18.07723	0.786701
di	53011.17004	6358.07
dv	57156.59765	7604.71
dj	43074.79818	13444.
dk	40276.76269	12644.
dh	43902.92989	14886.



SMC Cepheid

- a μ of 18.95 would give $\sim 60\text{kpc}$

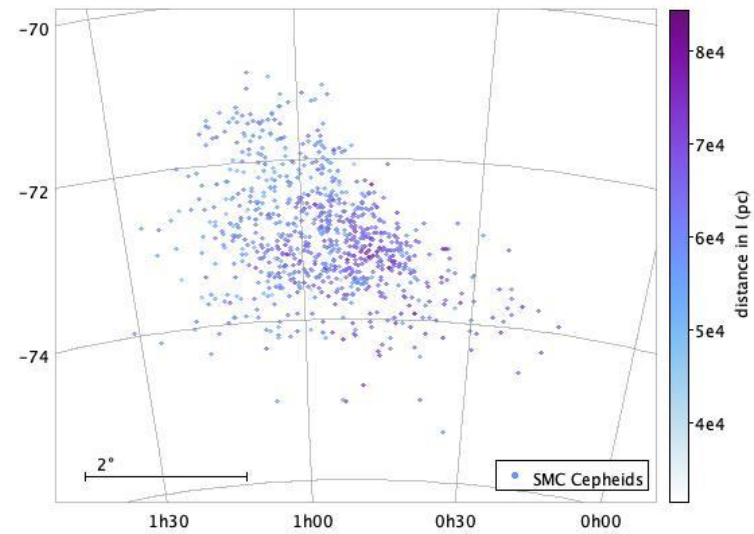
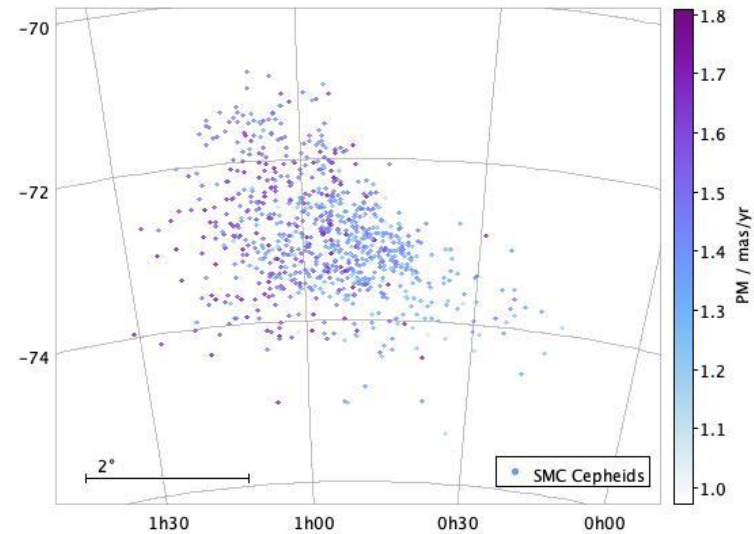
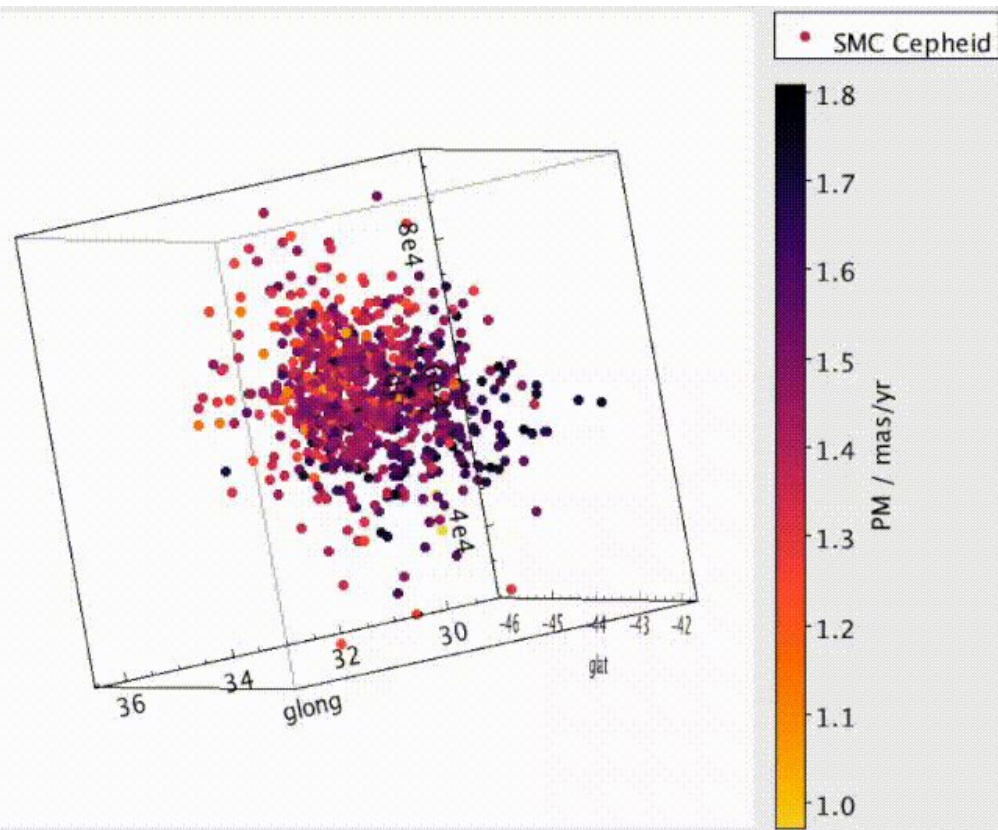


Name	Mean	STD
μ_i	18.92472	0.1669
μ_v	18.95235	0.209611
μ_j	19.05167	0.214307
μ_k	18.74823	0.358778
di	61126.06889	4694.69
dv	62014.03142	5980.73
dj	64930.85914	6433.8
dk	56961.43854	9542.8

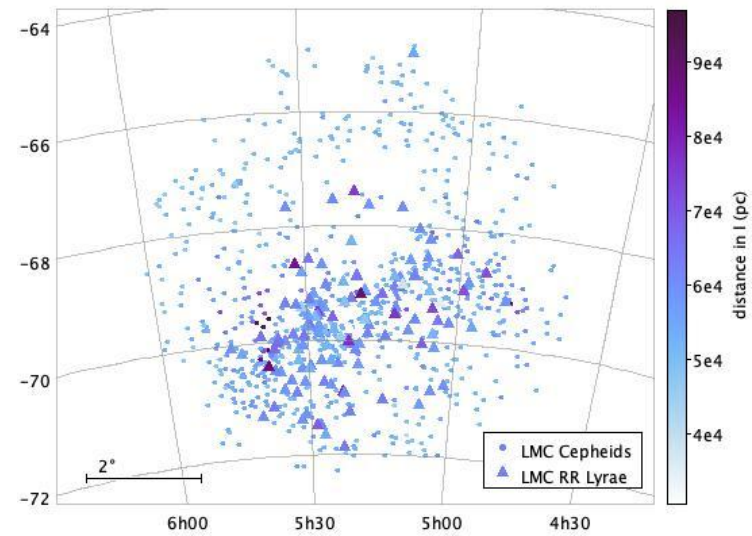
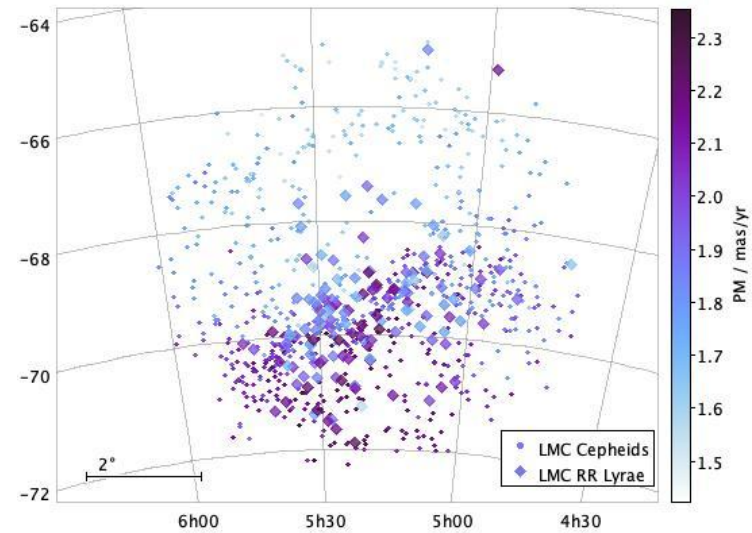
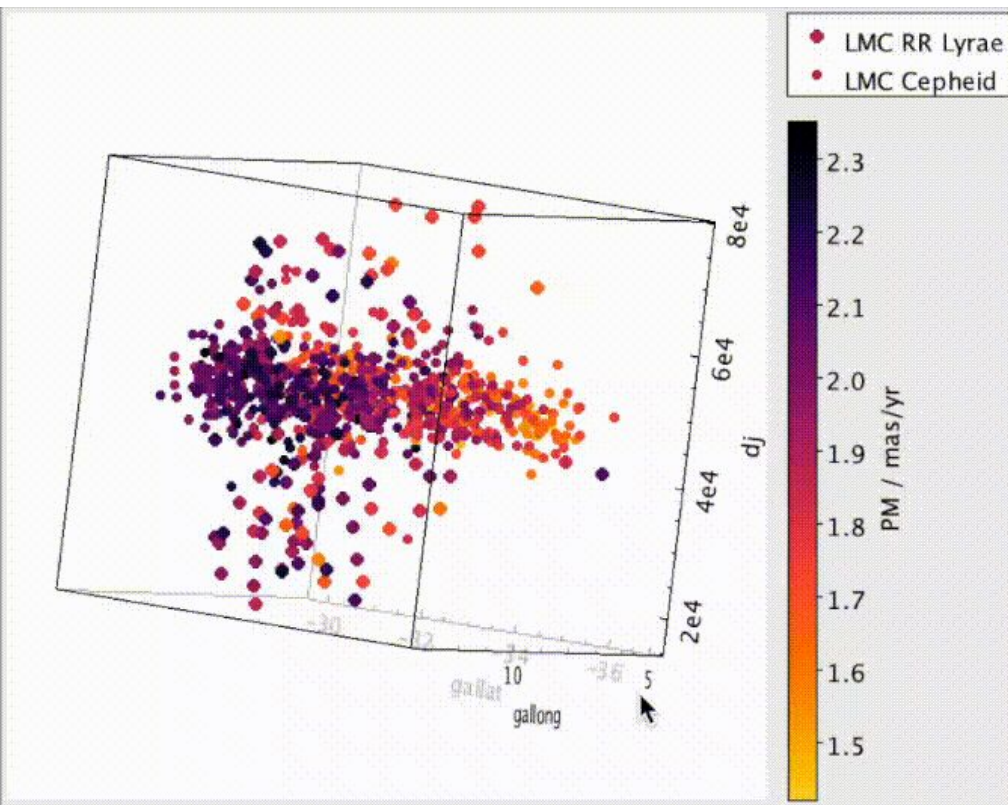
The cepheid relations with the OGLE/2mass periods and photometry yield $\sim$ literature values.



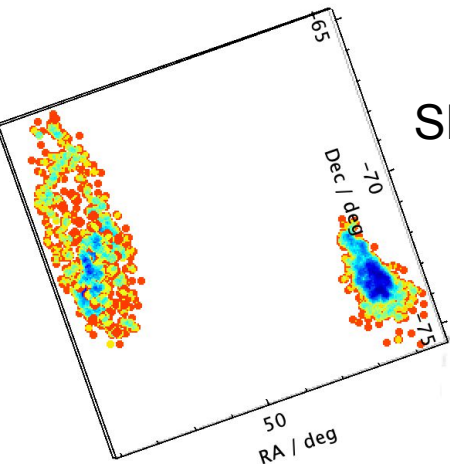
SMC



LMC



Results & Discussion



SMC distance: 61.2 ± 7.2 kpc

$d = 62.44$ kpc (Graczyk, et. al 2020)

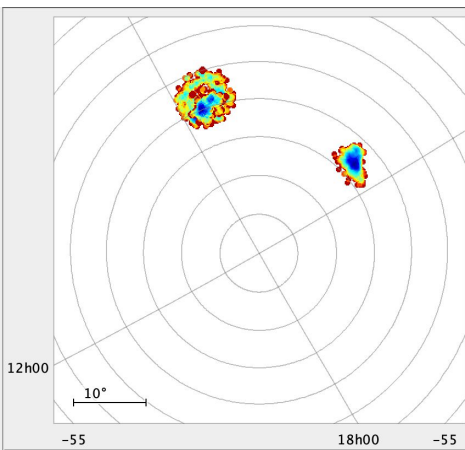
LMC distance: 51.2 ± 6 kpc

$d = 49.97$ kpc (Pietrzynski et al 2013)

The paper's period-luminosity relationship was decently accurate for the Large & Small Magellanic Clouds.

OGLE's database seemed amazing but oversaturation of background stars was apparent after the skyplot and PM cuts.

The reported error here is likely the dispersion of the stars, with the methods applied, rather than statistical error.

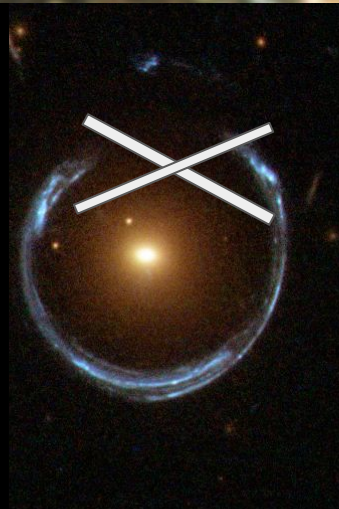
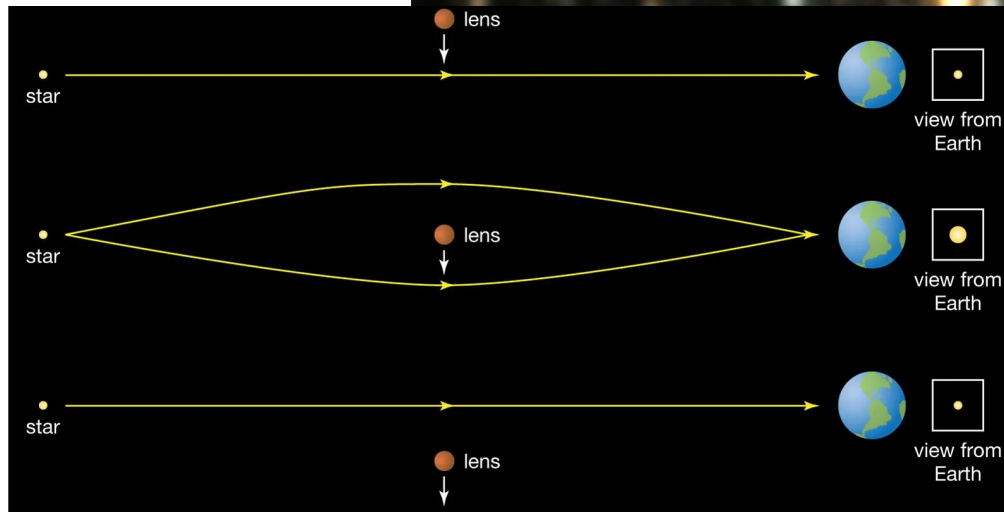
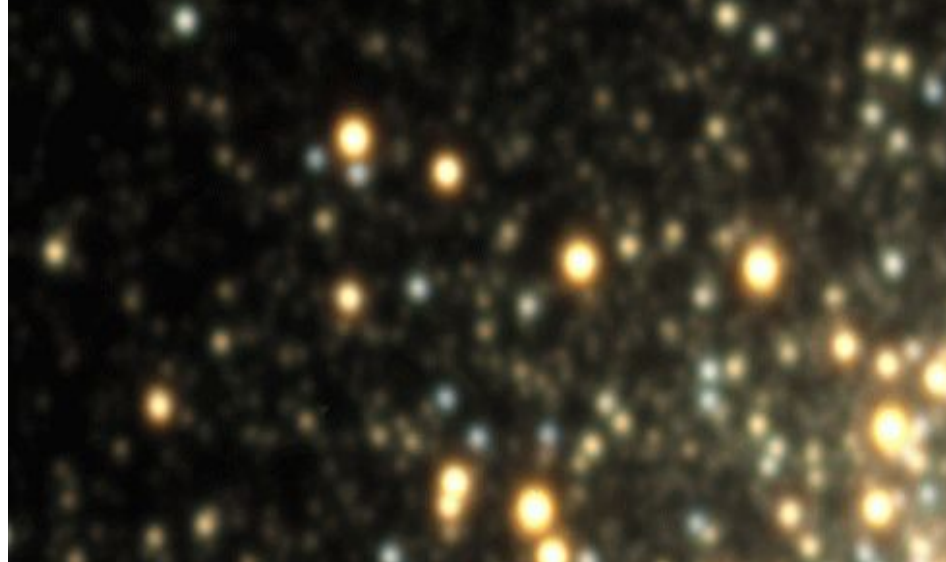


Bonus slide: Gravitational Lensing and variable stars

Long term survey of the southern sky, has been running every clear night since 1992.

Original purpose was to look for microlensing in dense fields of stars like the Galactic Bulge and Magellanic Clouds.

A change in brightness from intrinsic variability suggests a microlensing event caused by a dark matter candidate (MACHO), an exoplanet, or a brown dwarf passing between the observer and the star of interest, as the object's gravity temporarily magnifies the background star's light, warping the light curve in a non-repeating way.

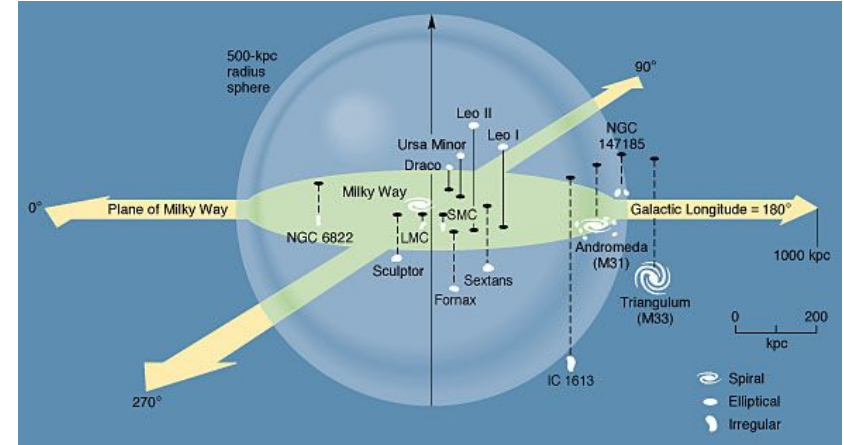


Bonus slide: OGLE distances of 54 Cepheids in dwarf galaxy IC 1613

	Distance mean	STD
d_LMv	717473.538871911	94548.22
d_LMi	708279.79296310 5	86850.64

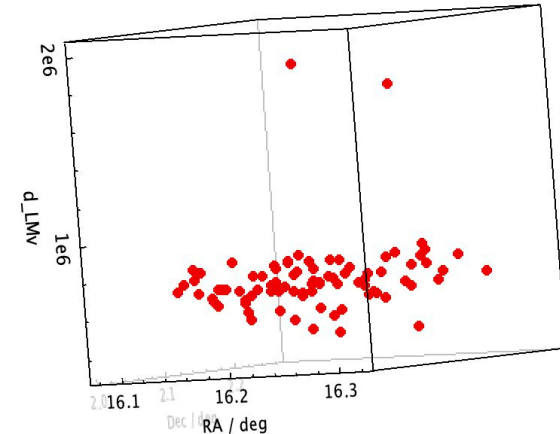
	Distance mean	STD
d_SMv	675389.33528201 17	88175.16
d_SMi	655975.61362849 42	79305.72

	Distance mean	STD
d_MMv	733880.0960763 406	95304.69
d_MMi	782892.2585497 8	95380.945

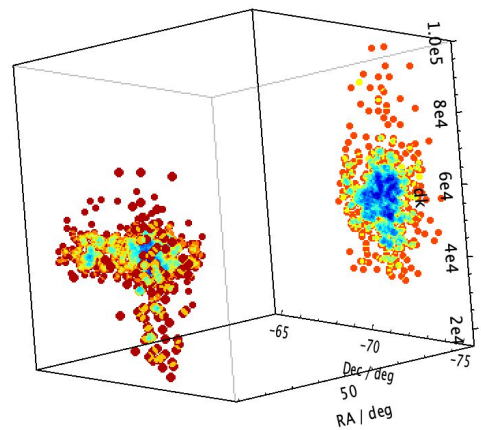
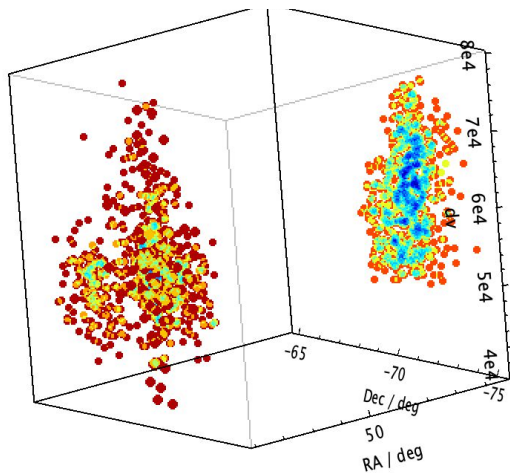
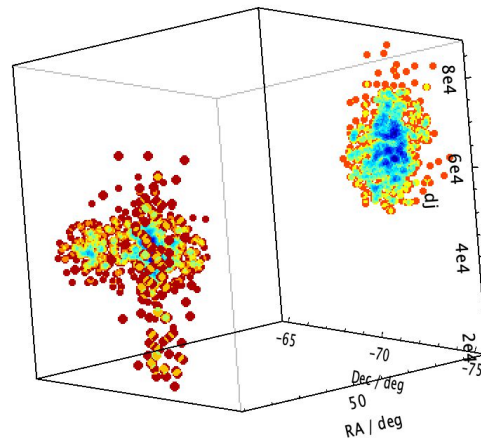
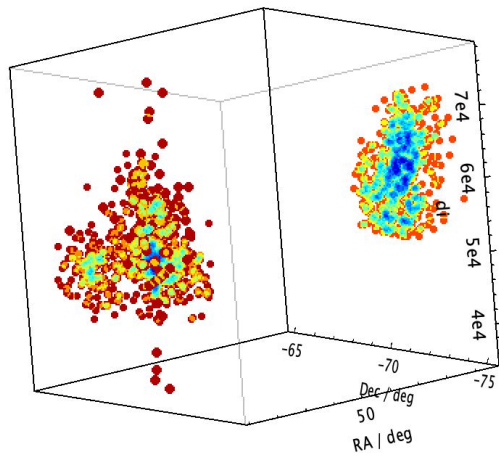


730±20 kpc (Karachentsev, Karachentseva, Makarov 2004)

None of the LMC, SMC, or MW Cepheid relations hit the mark here, but if we were to average the results of the LMC, SMC, and MW relations we would land roughly at this answer.



Bonus slide:



Different stars have different distance moduli ($m-M$) in different bands so each band has a different shape.

Local density plots shown in I (top left), V (bottom left), J (top right), and K (bottom right) bands.